LayoutAgent

LLM-Enhanced Agent-Based Generative Design for Outpatient Department Layouts

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Abstract. In Asia, the high population density and concentrated medical demand place persistent pressure on healthcare environments, requiring architectural design to balance efficiency and human-centered care. To address the low efficiency and limited data utilization in conventional design processes, this study proposes LayoutAgent, a generative design framework that integrates graph topology learning with Large Language Model (LLM)–enhanced agents for the automatic generation and optimization of outpatient department (OPD) layouts. The study first constructs a functional mask and spatial relation graph based on 120 OPD layout samples, from which a Graph Convolutional Network (GCN) learns spatial topology patterns. On this basis, an LLM-enhanced agent performs semantic reasoning and interactive topology adjustment. Experimental results demonstrate that the LLM-enhanced agent significantly improves topological prediction accuracy and achieves superior performance in flow efficiency and functional adjacency satisfaction compared with baseline methods. This research provides a novel pathway for the generative design of complex healthcare facilities and highlights the potential of LLMs in architectural semantic reasoning and design optimization.

Keywords. Outpatient department layout, large language model (LLM), multi-agent system, graph convolutional network (GCN), generative design

1. Introduction

In many Asian countries, the imbalance between medical resources and patient demand continues to intensify. The increasing population density and aging trend place heavy pressure on outpatient departments (OPDs), where design quality directly influences medical efficiency and patient experience. However, traditional architectural design methods largely rely on expert experience and manual iteration, making it difficult to efficiently explore the complex spatial relationships inherent in healthcare facilities and lacking objective, quantitative evaluation mechanisms (Rahbar et al., 2022). The

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introduction of generative design (GD) and artificial intelligence (AI) offers new possibilities for improving efficiency and methodological rigor in early-stage hospital layout planning.

Recent advances in graph neural networks (GNNs) and large language models (LLMs) have further expanded the potential of data-driven architectural reasoning. GNNs are effective in capturing spatial topology and relational structures, while LLMs exhibit strong capabilities in semantic understanding and contextual reasoning. However, existing data-driven approaches in healthcare architecture remain constrained by complex topological structures, diverse functional requirements, and limited availability of high-quality layout data, resulting in restricted generative performance (Bayraktar Sari & Jabi, 2024). Moreover, hospital design requires not only efficient layout generation but also coherent functional organization and human-centered spatial experiences, which have not been adequately addressed in current computational workflows (Jia et al., 2023).

To bridge this gap, this study proposes LayoutAgent, a generative design framework that integrates Graph Convolutional Network (GCN)–based topology learning with an LLM-enhanced agent. The system first learns spatial adjacency patterns from existing OPD layouts and subsequently employs the LLM agent for semantic reasoning, functional constraint checking, and interactive layout refinement. This framework improves the accuracy of topological prediction while enhancing the interpretability and adaptability of generative outcomes, offering a new paradigm for intelligent design assistance in healthcare architecture (Fig. 1).

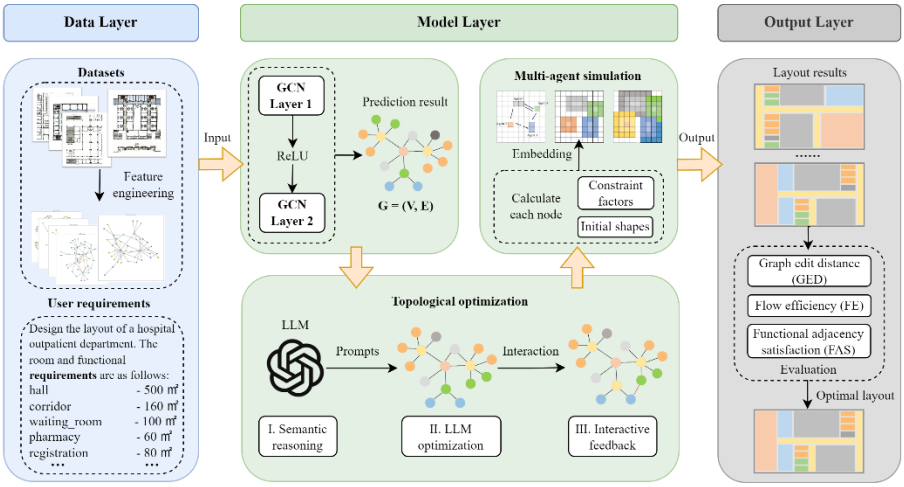


Figure 1. Research methodology procedure

2. Related Work

2.1. GENERATIVE DESIGN FOR HEALTHCARE ARCHITECTURE

Hospital buildings are among the most complex building types due to their functional diversity and tightly coupled spatial relationships (Jia et al., 2023). Owing to data

scarcity and topological complexity, generative design research in healthcare architecture has emerged later than in other domains (Bayraktar Sari & Jabi, 2024). Existing studies mainly focus on master planning and zoning optimization. For example, Cubukcuoglu et al. (2022) proposed a computational workflow integrating stacking, zoning, and routing for hospital design, though substantial manual intervention remains necessary. Zhao et al. (2021) applied generative adversarial networks (GANs) to emergency department layout generation, demonstrating the feasibility of deep learning approaches while largely neglecting functional semantics and topological logic. As a result, automated layout generation for healthcare facilities remains constrained by difficulties in learning complex functional and topological relationships.

2.2. GRAPH-BASED DEEP LEARNING FOR LAYOUT GENERATION

With the advancement of deep learning, graph-based methods have become a key direction in automated layout generation. Aalaei et al. (2023) introduced a graph-constrained GAN to map functional topologies to floor plan layouts, while Han et al. (2024) further achieved continuous image generation from adjacency graphs using the GRAPH2PIX model. Graph neural networks (GNNs) have also been combined with evolutionary algorithms for urban layout optimization (Li et al., 2025), and with agent-based modeling for hybrid layout generation (Rahbar et al., 2022). Other studies integrate graph algorithms with reinforcement learning to produce self-organized layouts (Wang et al., 2023; Zhang et al., 2024). However, most of these approaches target residential or generic building types and lack joint modeling of topology and semantic logic required for complex healthcare layouts.

2.3. MULTI-AGENT SYSTEMS AND LLMS IN GENERATIVE DESIGN

In architectural generative design (GD), agent-based approaches are widely used to simulate autonomous or semi-autonomous design behaviors. Early studies relied on rule-driven single-agent systems, such as Hua (2016), who employed graph-based inputs and morphological evolution to generate irregular layouts. With the rise of human-centered design, Wang et al. (2023) incorporated activity data into layout generation, emphasizing behavior-oriented design mechanisms. More recently, the integration of multi-agent systems with large language models (LLMs) has introduced a new paradigm. Qin et al. (2024) developed an LLM-driven intelligent design and optimization system capable of semantic-level reasoning and parameter adjustment, enabling explainable human–AI collaboration. Veloso and Krishnamurti (2023) further demonstrated the feasibility of generating spatial layouts through interactions among agents representing spatial units. These studies suggest that LLM-driven agents offer distinct advantages in semantic understanding, reasoning, and interactive optimization for complex spatial design tasks.

2.4. RESEARCH GAPS

Despite these advances, two major gaps remain. First, for highly complex building types such as hospitals, a unified framework that deeply integrates graph-based topology learning with layout generation is still lacking. Second, existing deep

learning–based methods often suffer from limited interpretability and insufficient interactive mechanisms, making them difficult to align with architects' semantic reasoning and design workflows. To address these gaps, this study proposes LayoutAgent, a generative design system that combines graph learning with LLM-enhanced agents to support an interpretable and interactive design process from semantic understanding to spatial generation.

3. Methodology

3.1. DATA PREPARATION AND PROCESSING

This study collected 120 outpatient department (OPD) floor plans from hospitals in China, the United States, and Japan, covering small- and medium-sized facilities as well as large general hospitals. These plans were used to construct the training dataset. During preprocessing, OPD spaces were categorized into 11 functional types, including entrance hall, corridor, consultation room, registration, and pharmacy, which were encoded using functional masks indexed from 0 to 10 (Fig. 2).

For each space, geometric attributes—including area, aspect ratio, and centroid coordinates—were extracted as node features. Spatial adjacency relationships were then identified to construct adjacency matrices, resulting in graph-structured data that integrate both functional and topological information. The final dataset contains approximately 5,000 nodes and 22,400 edges, which serves as the basis for subsequent topology learning and generation.

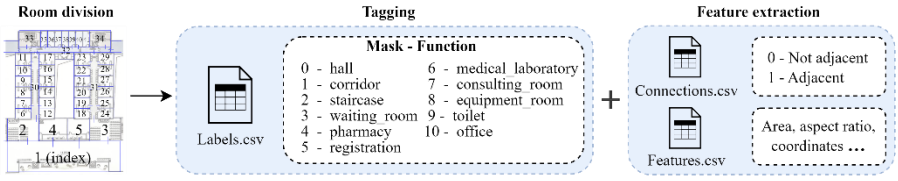


Figure 2. Feature engineering

3.2. TOPOLOGY GENERATION BASED ON GCN

To learn connectivity patterns among functional units in OPD layouts, a two-layer Graph Convolutional Network (GCN) was constructed. Each node represents a functional space, characterized by functional category and geometric attributes, including area, aspect ratio, and spatial coordinates.

Spatial topology is represented by a binary adjacency matrix, where an edge is labeled as 1 if two spaces are directly adjacent or have an explicit circulation relationship in the original layout, and 0 otherwise. This matrix is used both to constrain feature propagation and as supervision for topology prediction.

The first GCN layer aggregates features from neighboring nodes under graph constraints, followed by a ReLU activation. The second layer outputs connection probabilities between spatial units, enabling the prediction of OPD topological relationships. The model is trained using binary cross-entropy (BCE) loss to minimize discrepancies between predicted and ground-truth topologies, thereby capturing

functional organization patterns.

3.3. LLM-ENHANCED AGENT CONSTRUCTION AND TOPOLOGY OPTIMIZATION

After the GCN generates an initial spatial topology, an LLM-enhanced agent system (LayoutAgent) is introduced to perform semantic optimization and interactive refinement of the topology (Fig. 3). The LLM employed in this study is ChatGPT (GPT-4, OpenAI, 2024), with a temperature setting of 0.7.

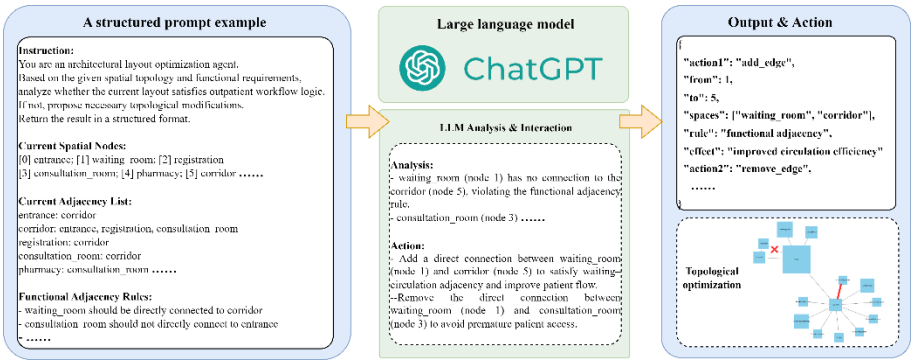


Figure 3. LLM topology optimization process

The agent system consists of three modules: (1) Information Parsing Module: the LLM parses the generated layout, identifies potential logical conflicts, and detects functional inconsistencies; (2) Semantic Reasoning Module: based on a structured prompt framework and incorporating medical-building semantics and design specifications, the module performs topology optimization (Table 1); (3) Interactive Feedback Module: designers may use natural language to interact with the agent to steer the optimization direction, enabling a human–AI co-creation workflow. Leveraging the LLM's semantic understanding and reasoning capabilities, the system improves the rationality and interpretability of the generated topology while maintaining efficiency.

Table 1. Prompt Framework for LayoutAgent

Step	Prompt Component	Instructions
1	Role Definition	Define the LLM as a semantic agent for medical layout topology, responsible for interpreting GCN-generated graphs and identifying logical inconsistencies, without creating new spaces or changing functional requirements.
2	Task Goals	Perform functional semantic checking, adjacency and circulation evaluation, and provide interpretable topology optimization suggestions supporting stepwise interaction.

3	Reasoning Constraints	Follow medical workflow logic, avoid infeasible adjacencies, and prioritize hard constraints over soft preferences to ensure functional and circulation correctness.
4	Callable Commands	Restrict actions to predefined operations (semantic checking, adjacency evaluation, topology optimization, and explanation), avoiding undefined behaviors.
5	Response Format	Output structured JSON including detected issues, proposed actions, reasoning, and optimized adjacencies, ensuring machine readability and downstream conversion.
6	Dialogue Mechanism	Maintain dialogue context (e.g., design preferences and prior adjustments) to support iterative and consistent human–agent topology refinement.

3.4. LAYOUT GENERATION VIA MULTI-AGENT SIMULATION

After topology determination, geometric layout generation is performed by multi-agent simulation. Each agent represents a spatial unit and adjusts its position through an attraction–repulsion mechanism: topologically connected spaces attract each other to maintain adjacency, while functionally incompatible or scale-mismatched spaces generate repulsive forces to prevent unreasonable configurations (Fig. 4).

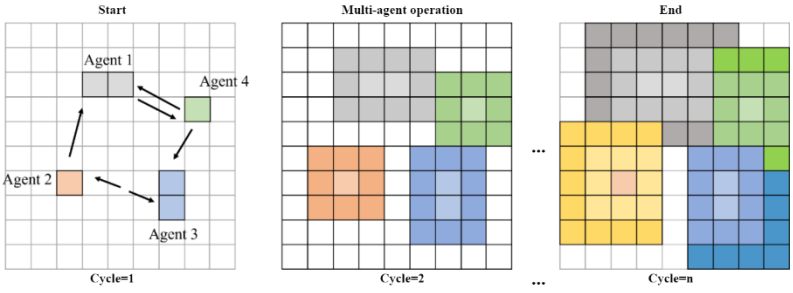


Figure 4. Agent-based attraction–repulsion mechanism and growth rules

Functional constraint factors are embedded in node attributes, enabling agents to simultaneously consider topological relationships, functional logic, and geometric constraints during layout formation. The LLM-enhanced module further provides semantic interpretation and dynamic decision-making, allowing rule conflicts or layout anomalies to be resolved through language-based reasoning. Guided by predefined growth rules, the layout is iteratively expanded and validated against adjacency consistency after each iteration until all constraints are satisfied, resulting in layouts that align with medical workflows and human-centered spatial logic (Fig. 4).

4. Experiments and Results

4.1. MODEL TRAINING AND TOPOLOGY GENERATION

The graph generation model was trained for 2000 epochs on an NVIDIA GeForce RTX 4080 GPU, with an initial learning rate of $\eta = 0.001$. Data augmentation was applied through mirroring, rotation, and random noise perturbation, and the dataset was split into non-overlapping training and validation sets. After multiple rounds of hyperparameter tuning, this learning rate yielded the best performance, and early stopping was triggered at epoch 1560. In the binary classification task for topological connectivity prediction, the model achieved an accuracy of 85.3%. As shown in Fig. 5, the loss curves of both the training and validation sets decrease steadily and converge at approximately 0.5, indicating satisfactory generalization performance.

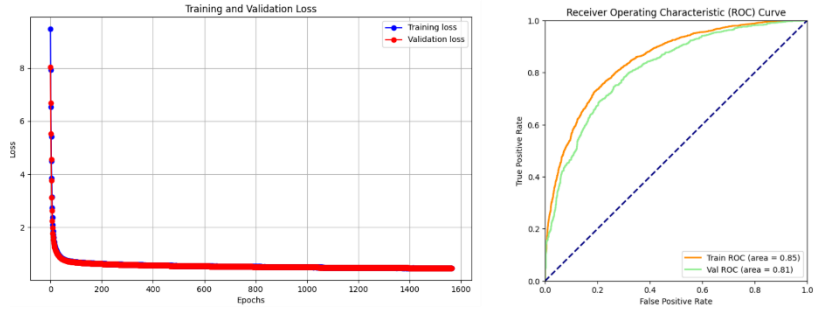


Figure 5. GCN Model training loss and AUC, ROC curves

Ideally, a Receiver Operating Characteristic (ROC) curve closer to the upper-left corner indicates stronger discriminative capability, while an Area Under the Curve (AUC) value approaching 1 reflects better classification performance. In this study, AUC values reached 0.85 (training) and 0.81 (validation), indicating stable discriminative capability and robustness in spatial adjacency prediction (Fig. 5). The model consistently generated structurally stable and interpretable spatial topologies under varying outpatient functional requirements (Fig. 6).

4.2. AGENT OPTIMIZATION AND LAYOUT GENERATION

Under identical functional requirements, three generation strategies were compared: (1) GCN-only, topology generated solely by the GCN model; (2) LLM-only, topology generated solely by the LLM-based agent; (3) GCN+LLM-Agent, topology generated by the GCN and refined via LLM-enhanced interaction (proposed method).

In the experimental group (GCN+LLM-Agent), the topological results produced by the GCN were input into the LLM-enhanced agent system, where the agent conducted semantic reasoning and topological refinement. After optimization by the LLM-enhanced agent, the accuracy of topology-relation prediction improved from 77.2% to 89.5%. Each resulting topology was then assigned a Functional Constraint Coefficient, followed by multi-agent simulation for geometric generation. During simulation, agents adjusted spatial positions dynamically according to attraction–repulsion mechanisms and rule constraints, producing layouts that maintained

topological logic and geometric coherence (Fig. 6).

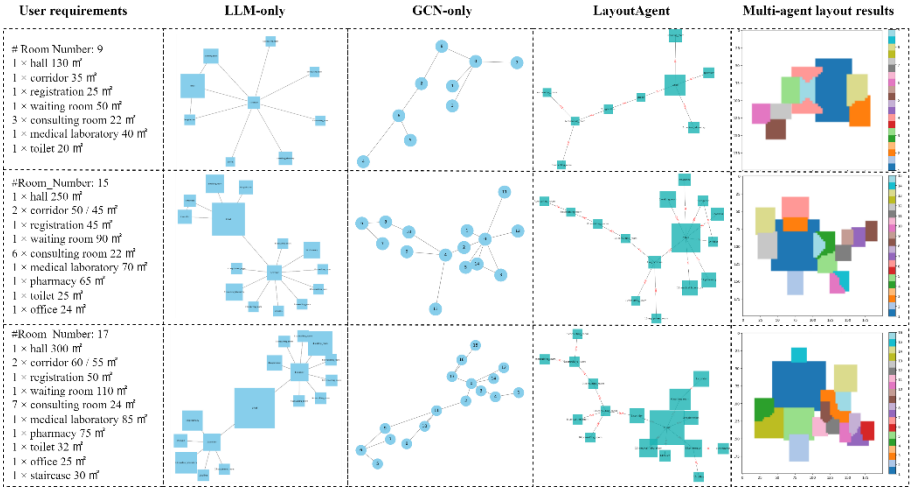


Figure 6. Topology optimization and layout generation

4.3. QUANTITATIVE COMPARISON AND EXPERT EVALUATION

To assess the rationality of spatial organization across different methods, three complementary metrics were employed: Graph Edit Distance (GED), Flow Efficiency (FE), and Functional Adjacency Satisfaction (FAS). GED measures the minimum topological editing cost between generated layouts and reference outpatient layouts based on adjacency graphs. FE evaluates circulation compactness and accessibility by comparing the shortest topological path from the entrance to key functional spaces with their corresponding geometric distances. FAS quantifies the proportion of spatial adjacencies that conform to core outpatient workflow logic.

As summarized in Table 2, LayoutAgent outperformed the baseline methods across all three metrics, achieving a 32% reduction in GED, a 15% improvement in FE, and a 23% increase in FAS. These results indicate that the integration of an LLM-enhanced agent enables coordinated optimization of topological consistency, circulation efficiency, and functional logic.

Table 2. Comparative analysis results

Method	GED (Lower is better)	FE (Higher is better)	FAS (Higher is better)
LLM-only	0.412	0.684	0.713
GCN-only	0.365	0.711	0.720
LayoutAgent	0.248	0.817	0.884

In addition, 36 architects and healthcare design experts with experience in hospital or public building projects were invited to conduct a subjective evaluation. Using a five-point Likert scale, layouts were assessed from three dimensions: overall coherence, functional rationality, and spatial flexibility. Functional rationality focused

on outpatient workflow completeness, relative positioning of key spaces, and circulation clarity. The results show that LayoutAgent achieved an average score of 4.57/5, significantly outperforming GCN-only (3.86) and LLM-only (3.12). Experts generally agreed that the generated layouts are suitable for early-stage outpatient design evaluation and optimization, and that the integration of graph learning and semantic reasoning improves both circulation efficiency and functional coherence.

5. Discussion

This study verifies the potential of integrating Graph Neural Networks (GNNs) and Large Language Models (LLMs) in generative healthcare design, yet some limitations remain. Although semantic indicators such as FE and FAS were introduced, they still fall short of fully capturing human-centered factors. The performance of LayoutAgent also depends on LLM prompt design and semantic accuracy—improper prompts may misguide optimization. Moreover, given the diversity of real-world hospital layouts in scale and organizational patterns, further validation is required to assess generalization in practical projects.

Despite these challenges, the integration of LLMs significantly enhances interpretability and designer interaction, enabling semantic understanding of model decisions. This suggests that future architectural agents should combine understanding and reasoning abilities, moving beyond single-mode pattern recognition.

6. Conclusion

This study proposes LayoutAgent, a generative design framework that integrates graph-topological learning with Large Language Model (LLM)–enhanced agents, providing a novel technical pathway for the automatic generation of complex healthcare floor layouts. The contributions of this research are threefold: (1) the development of a generative design framework combining graph learning with LLM-based semantic reasoning; (2) the introduction of an LLM-enhanced agent enabling interpretable and interactive spatial optimization; and (3) the construction of a graph-structured dataset and evaluation system tailored to healthcare architecture. Experimental results demonstrate that LayoutAgent not only improves the efficiency and logical consistency of outpatient layout design but also extends the applicability of generative design to complex architectural layouts.

Future work will extend the framework to larger healthcare and public-building datasets and incorporate multimodal inputs and 3D representations to better support human-centered, AI-assisted design.

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